

MODULATION & PERCEPTION OF PAIN



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LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- ✓ Define pain modulation and pain perception.
- ✓ Describe the major pain modulation pathways including ascending & descending pathways.
- ✓ Identify key structures involved in descending pain modulation including periaqueductal gray(PAG), rostral ventromedial medulla(RVM) & dorsal horn of the spinal cord.
- ✓ Differentiate between inhibitory & facilitatory pain modulation.
- ✓ Describe how psychological factors like attention, anxiety, expectation, emotion affect pain perception.
- ✓ Differentiate between pain modulation & pain perception.
- ✓ Explain the role of endogenous opioids, serotonin & noradrenaline in modulating nociceptive signaling.

WHAT IS PAIN?

- Pain is an unpleasant sensory & emotional experience linked to actual or potential tissue damage.
- According to *International Association for the Study of Pain*, pain is a deeply subjective experience shaped by biological, psychological & social factors.
- Therefore, pain is both a physical sensation & an emotional reaction.
- It acts as a warning sign to protect the body from harm.

STAGES OF PAIN PROCESSING & PERCEPTION

- Pain is not simply the electrical signal arriving from the injured tissue rather it is a sensory & emotional experience whose perception can be modified at multiple levels of the nervous system.
- The central nervous system (CNS) can amplify, reduce or modify the incoming signal.
- Pain processing can be understood in four major stages:

Stimulus => Transduction => Transmission => Modulation => Perception

TRANSDUCTION

- Specialized nerve endings called **nociceptors** convert physical, thermal or chemical threats into electrical nerve signals.

TRANSMISSION

- Signals travel from peripheral receptors via **fast A-delta fibers** (for sharp pain) & **slow C fibers** (for dull ache) to the dorsal horn of the spinal cord & then ascend to the brain.

MODULATION

- The nervous system can turn pain signals up or down.
- These signals are modified within the nervous system i.e. **inhibited** or **facilitated**.
- Descending pathways from the brain release natural chemicals like **endogenous opioids** to inhibit pain signals at the spinal cord level.

PERCEPTION

- The brain interprets the final signal. Widespread areas including the **somatosensory cortex, insula & limbic system** create the conscious awareness, location & emotional distress of pain.
- Therefore, pain perception involves a distributed network rather than a single pain center.

Transduction

Harmful stimuli (mechanical, thermal, or chemical) are converted into electrical signals in nociceptors

Transmission

These electrical signals are transmitted via peripheral nerves to the spinal cord and brain

Modulation

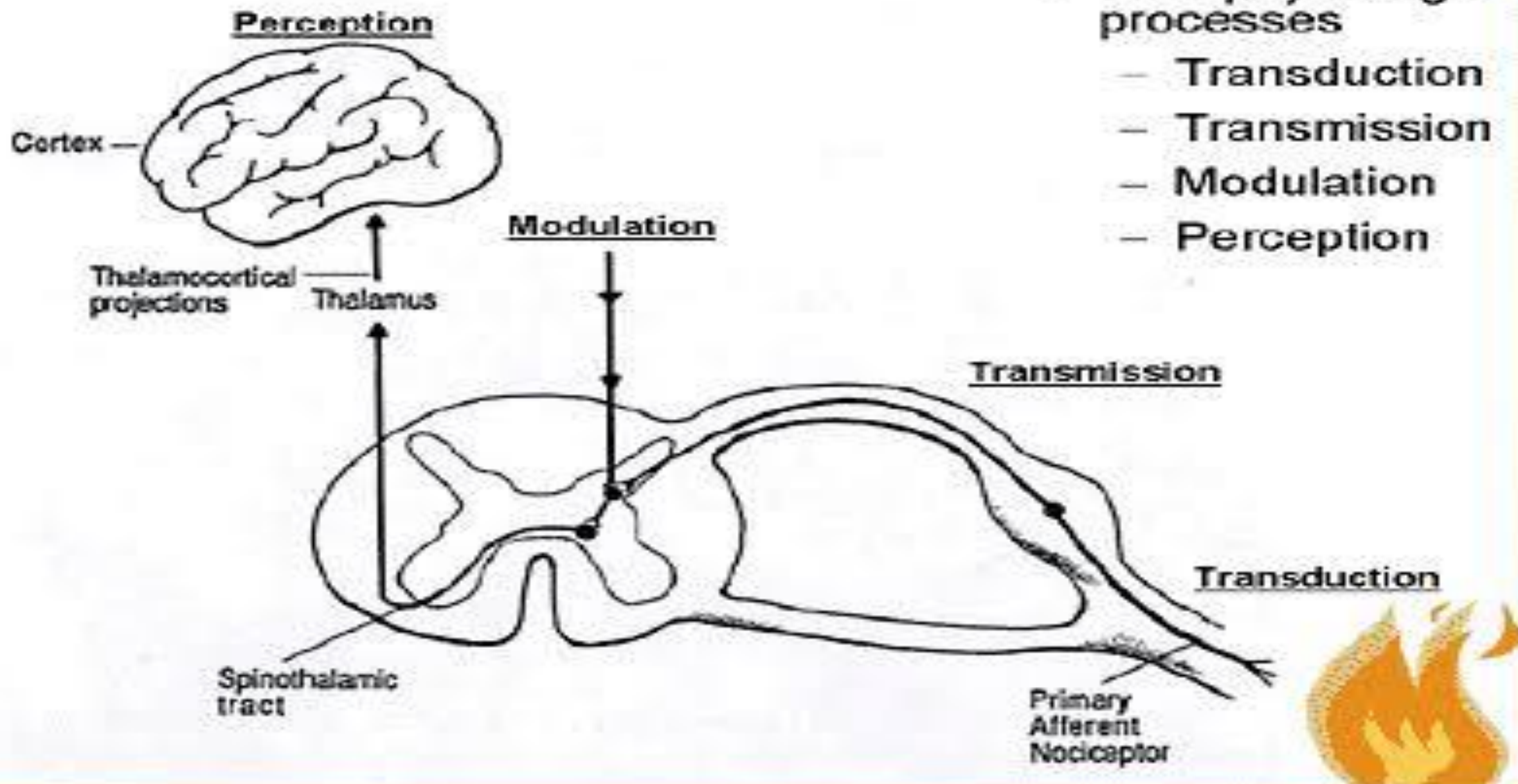
The nervous system can modulate pain signals through descending pathways, either amplifying or inhibiting the sensation of pain

Perception

The brain processes and interprets these signals as pain. This is influenced by emotional and psychological factors

♦ Four physiologic processes

- Transduction
- Transmission
- Modulation
- Perception



ROLE OF THALAMUS IN THE PATH TO CONSCIOUSNESS

- The thalamus acts as an important **relay station**.
- Thalamus is a structure deep inside the brain that acts as a central switching station in the journey of pain signals.
- The thalamus organizes the signals & forwards them to the surface of the brain.
- From the thalamus, information reaches several cortical & subcortical areas of the brain.

WHERE DOES PAIN BECOME CONSCIOUS?

- Pain becomes conscious in the **cerebral cortex**, the outermost layer of the brain.
- While pain signals travel through multiple pathways, they do not turn into a conscious felt experience until they reach this highly developed area of the brain.

BRAIN REGIONS THAT PROCESS PAIN

- Once the signal reaches the cortex, different parts of the brain work together to create the full experience of pain.
- The important areas in the processing of pain are:

PRIMARY SOMATOSENSORY CORTEX

- Primary somatosensory cortex decides where the pain is located & how intense it feels.

INSULAR CORTEX

- It integrates the sensory information & gives internal awareness of the pain.
- Evaluates the internal threat level to the body.

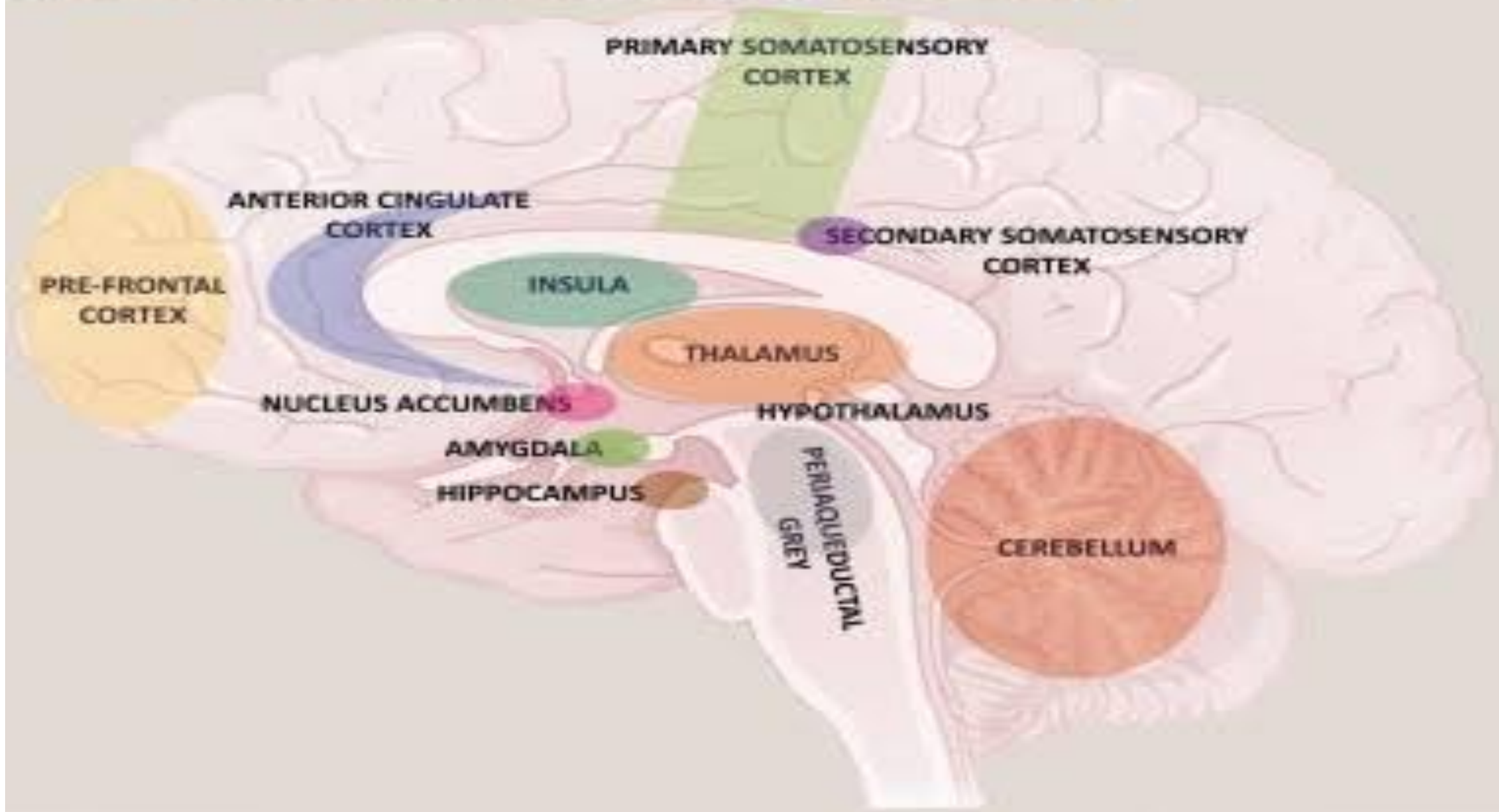
ANTERIOR CINGULATE CORTEX

- It processes the emotional distress & unpleasantness of the pain.

OTHER LIMBIC AREAS (PREFRONTAL CORTEX)

- Handles the thoughts, memories & attention towards the pain.

Areas of the brain involved with pain processing



PAIN MODULATION

Pain modulation is the process by which the nervous system alters the intensity or quality of nociceptive signals before they produce the final conscious experience of pain.

□ Pain modulation may *decrease pain* by *inhibition*

Or

□ It may *increase pain* through *facilitation*.

□ Hence, pain modulation is the CNS control of nociceptive transmission.

WHY IS PAIN MODULATION NECESSARY?

- The nervous system does not transmit every painful signal to consciousness with equal intensity.
- Pain modulation allows the body to:
 1. Suppress unnecessary pain.
 2. Focus attention during emergencies.
 3. Respond appropriately to injury.
 4. Prevent excessive nociceptive activity.
 5. Adapt pain responses according to situation.

CLINICAL EXAMPLE

A person having a road accident may sustain a significant injury but initially report surprisingly little or no pain because strong descending inhibitory mechanism can temporarily suppress pain. Later when the danger has passed, pain may become prominent.

PAIN MODULATION PATHWAYS

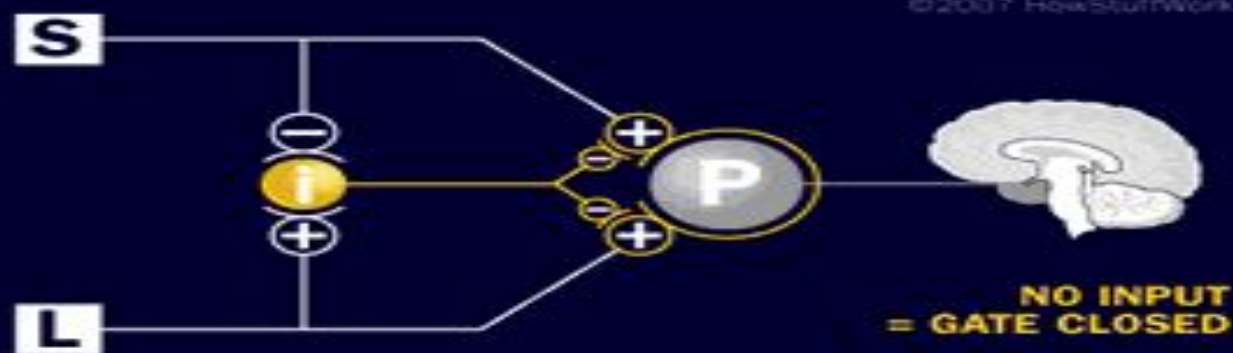
- Pain modulation occurs at four main sites along the nervous system pathways, where pain signals can be either dampened(*inhibited*) or amplified(*facilitated*).
- The primary site where the body naturally blocks pain before it reaches consciousness is the *dorsal horn of the spinal cord*.
- Specific anatomical sites where pain modulation takes place are:

PERIPHERAL RECEPTORS(TISSUE LEVEL)

- Modulation begins right at the site of injury.
- When tissue is damaged, it releases inflammatory mediators(prostaglandins, histamine, bradykinin) which makes pain receptors more sensitive.
- Naturally, the body releases anti inflammatory compounds to slow down signal generation.
- Medically, non-steroidal anti-inflammatory drugs(NSAIDs) like ibuprofen target this site by blocking inflammatory chemicals.

How Pain Works The Melzack-Wall Pain Gate

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S Small Nerve Fibers

L Large Nerve Fibers

i Inhibitory Neuron

P Projection Cells

DORSAL HORN OF THE SPINAL CORD (THE GATE)

- This is the most critical local site for pain regulation which can be explained by the **Gate Control Theory** of pain.
- Nerves that detect the safe touch like rubbing or pressure can physically block(**close the gate**) on the smaller nerves carrying pain signals.
- Local neurons(interneurons)in the spinal cord release inhibitory chemicals like GABA, enkephalins, glycine to stop pain signal from travelling up to the brain.

BRAINSTEM (DESCENDING PAIN-CONTROL CENTER)

- The brain can actively send signals downward to suppress incoming pain(**descending inhibitory pain pathway**).
- In this pathway, brain send signals downward towards the spinal cord to inhibit nociceptive transmission.

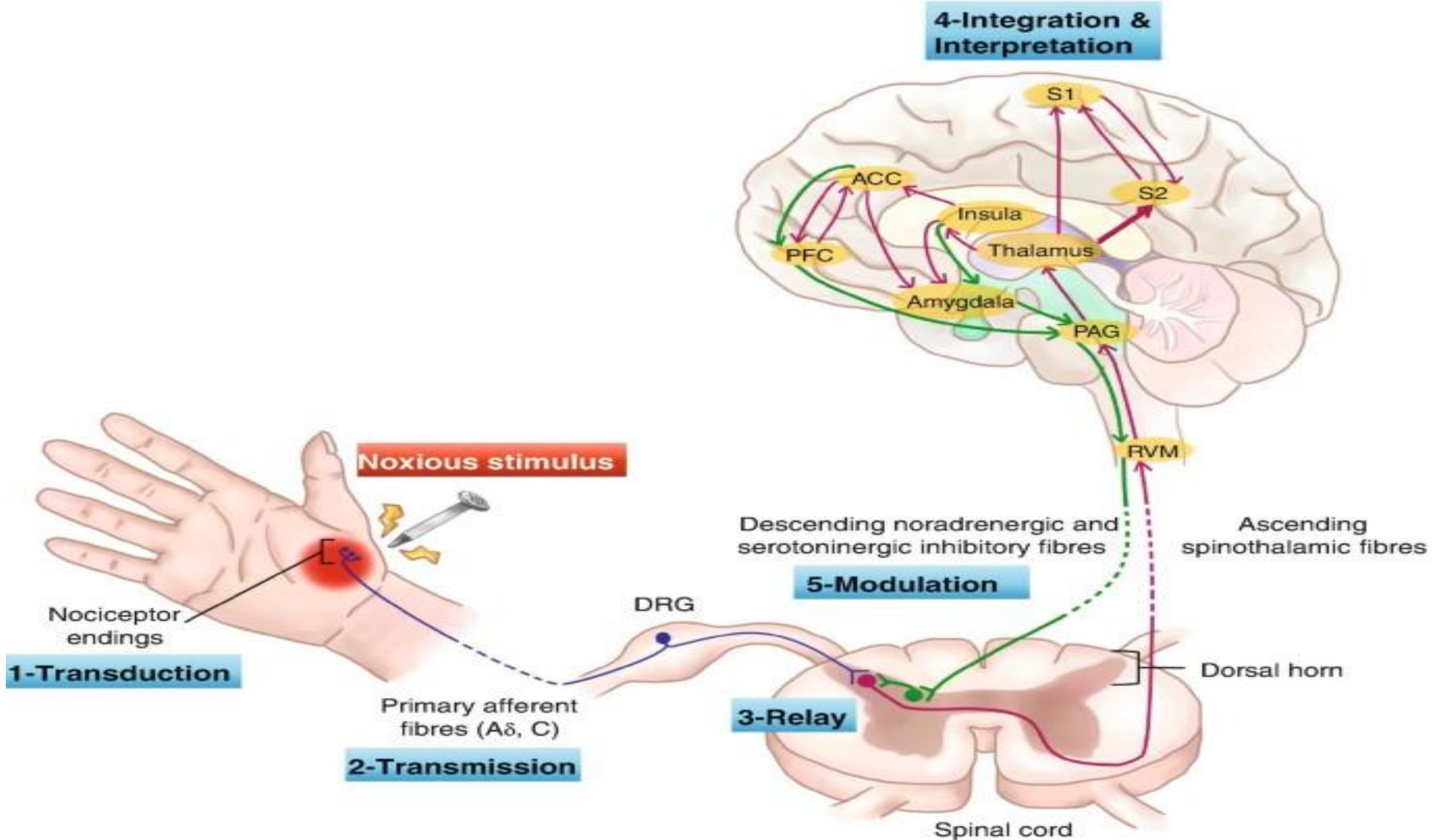
Descending inhibitory pathway relies on two primary brainstem structures namely:

1. PERIAQUEDUCTAL GRAY (PAG)

- Periaqueductal gray(PAG) is a region in the brainstem that surrounds the cerebral aqueduct.
- It is an important *supra-spinal center for pain inhibition* & activates under extreme stress or survival conditions effectively shutting off pain transmission.
- PAG activates *medullary pain-inhibitory centers* which ultimately produces inhibition of nociceptive transmission in the spinal cord.

2. NUCLEUS RAPHE MAGNUS/MEDULLARY RAPHE REGION

- It is a structure located in the medulla & is an important component of descending pain-control system.
- It send signals down the spinal cord to release neurotransmitters which act as natural painkillers like *serotonin* & *norepinephrine*.



DESCENDING PAIN MODULATORY SYSTEM

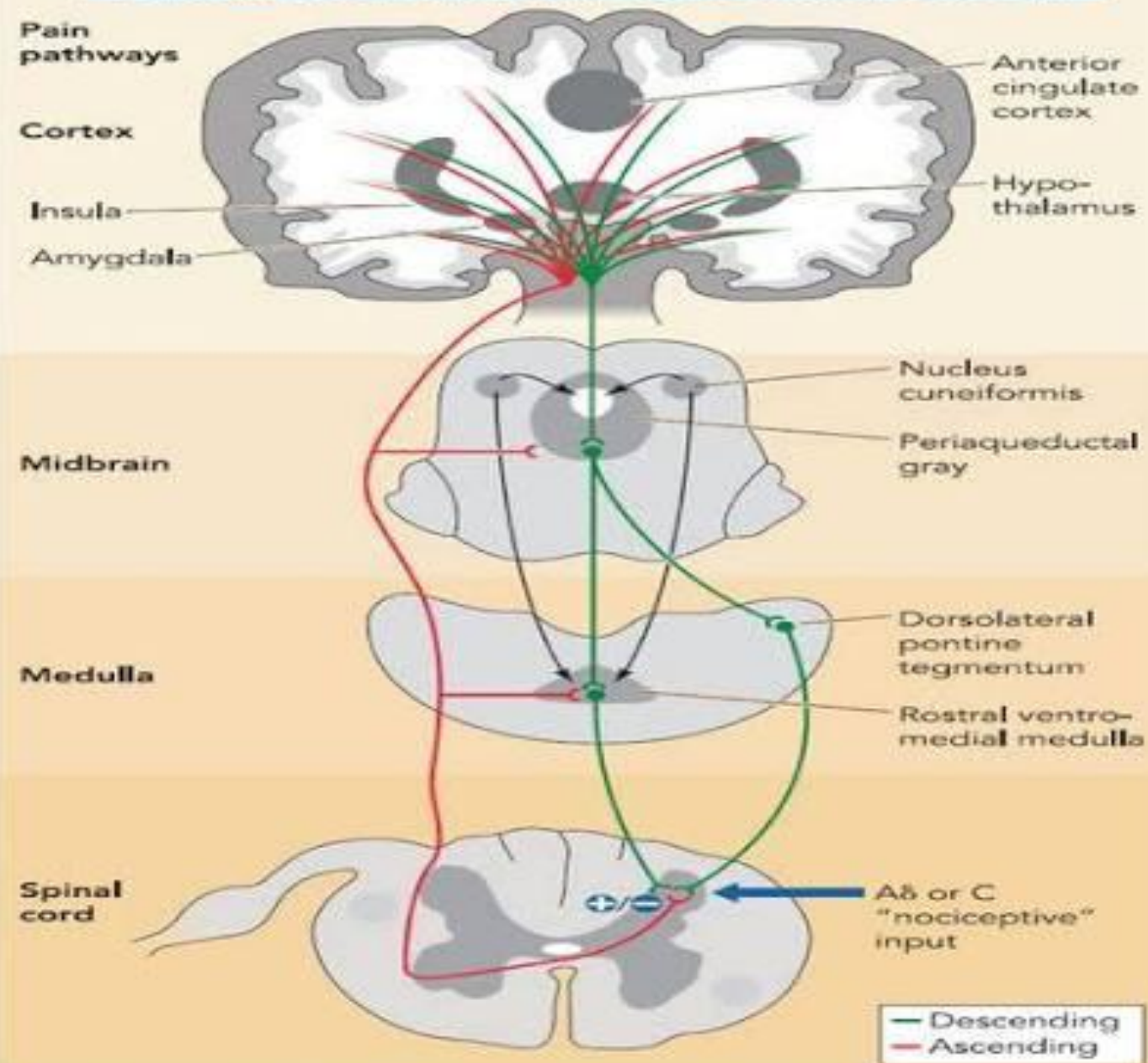


FIGURE 3. The descending pain modulatory system
+/- indicates both pro- and anti-nociceptive influences, respectively.

CEREBRAL CORTEX & LIMBIC SYSTEM (HIGHER BRAIN)

- Feeling of pain can be directly influenced by psychological factors like anxiety, distraction or focus can raise or lower the final perception of pain.

COGNITIVE MODULATION

- The prefrontal cortex & the amygdala can trigger the brainstem to release *endorphins*.
- For example, extreme focus, placebo effects or psychological distraction can significantly lower perceived pain.

HOW DOES INHIBITION OCCUR?

Descending inhibition can act at the presynaptic & post-synaptic levels.

PRESYNAPTIC INHIBITION

Presynaptic inhibition reduces neurotransmitter release from primary nociceptive afferents i.e. reduced glutamate/substance P release.

POSTSYNAPTIC INHIBITION

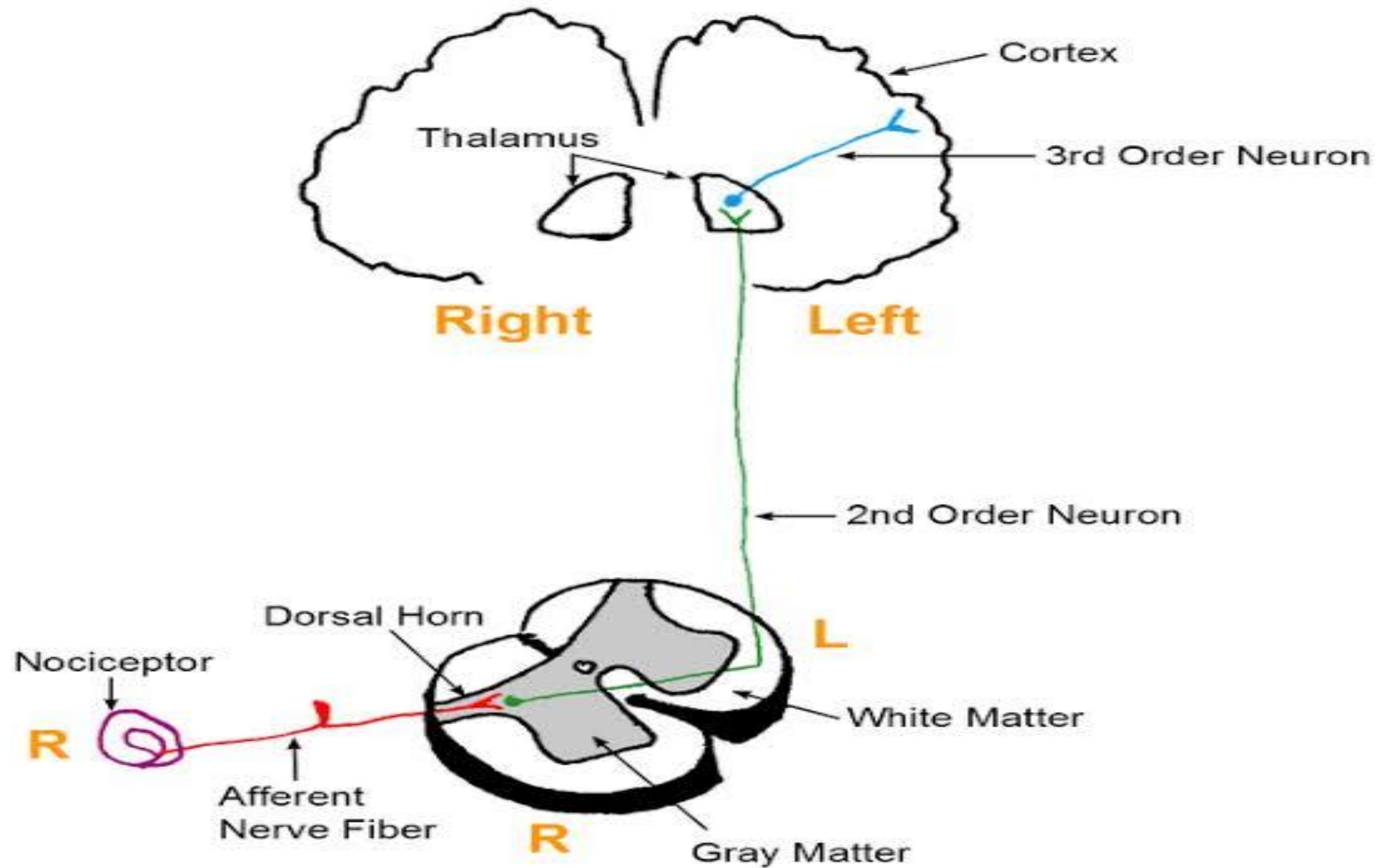
Inhibition at post synaptic level makes the second order neurons less excitable.

EFFECT

This presynaptic & postsynaptic inhibition causes:

Less nociceptive information reaches ascending pathways

□ Less pain perception



PAIN MODULATION CHANGES PAIN PERCEPTION

- Pain modulation is the modification of nociceptive signals whereas, pain perception is the final conscious experience of that pain.
- In other words, modulation changes the signal & perception reflects that modified signal.
- Modulation directly shapes perception by deciding how strong, weak or urgent the final pain signal feels in the brain.

How pain modulation changes pain perception?

- Think pain pathway as a volume control system in which peripheral injury produces the incoming signal, spinal cord then processes & modifies that signal & descending pathways can either **turn the volume down(inhibition)** or **turn the volume up(facilitation)**.
- Finally brain interprets the final signal as **PAIN PERCEPTION**.

INHIBITION (TURNING DOWN)

- Descending pathways from the brain i.e. PAG-RVM pathway release natural painkillers like endorphins to block signals at the spinal cord.
- This makes a person perceive less pain.

FACILITATION (TURNING UP)

- The nervous system can amplify signals through central sensitization in which normal touch or mild pressure can feel extremely painful.

GATE CONTROL THEORY

- Non-painful stimuli like rubbing a bumped elbow activate large nerve fibers that close the spinal gate by blocking signal transmission through smaller nerves & reducing the pain signals that reach the brain for perception.

PSYCHOLOGICAL INFLUENCE

- Stress, anxiety or negative expectations open the gate & hence amplify pain perception.
- Calmness, distraction or positive expectations trigger inhibition to decrease perception.
- For example, two individuals may have similar tissue injury but experience different levels of pain because pain is influenced by descending inhibition, descending facilitation, attention, anxiety, fear previous experience & emotional state.

PAIN MODULATION VS PAIN PERCEPTION

MODULATION

- ❑ Pain modulation is the nervous system's process of turning down or turning up pain signals.
- ❑ Modulation occur in the spinal cord & brain.
- ❑ Modulation can inhibit or facilitate signals.
- ❑ Modulation may occur before conscious awareness.
- ❑ Modulation is influenced by descending pathways.

PERCEPTION

- ❑ Pain perception is the actual conscious awareness or feeling of that pain.
- ❑ Perception mainly involves higher CNS processing.
- ❑ Produces awareness of pain.
- ❑ Perception requires conscious processing.
- ❑ Perception is influenced by sensory & emotional factors.

CHEMICAL MESSENGERS IN PAIN MODULATION

The nervous system uses specific neurotransmitters to alter pain signals:

- 1. *ENDORPHINS & ENKEPHALINS*:** body's natural opioids that bind to receptors to block pain transmission.
- 2. *SEROTONIN & NOREPINEPHRINE*:** neurotransmitters utilized by the descending pathway to inhibit pain signals in the spinal cord.
- 3. *SUBSTANCE P & GLUTAMATE*:** neurotransmitters in the ascending pathways that amplify & drive the pain signal forward(facilitation).

ENDOGENOUS OPIOIDS

□ Endogenous opioids are:

natural pain-relieving chemicals made by the body.

□ They attach to specific cell receptors in the nervous system to block pain signals, reduce stress & create comfort.

TYPES OF ENDOGENOUS OPIOIDS

1. **ENDORPHINS:** found mostly in the brain; blocks pain & boost mood during stress or hard exercise.
2. **ENKEPHALINS:** concentrated in the spinal cord; stops pain messages before they reach the brain.
3. **DYNORPHINS:** regulate heavy pain responses, but high amounts can cause unpleasant feelings.

How do endogenous opioids reduce pain?

- Endogenous opioids(endorphins & enkephalins) reduce pain by binding to specific G protein-coupled receptors(i.e. *mu, delta, kappa receptors*) in the brain & spinal cord.
- *Presynaptically*, endogenous opioids *close calcium channels* to stop calcium from entering the cell resulting in reduced release of pain signaling chemicals(glutamate or substance P) & thus effectively blocks ascending pain signals from reaching the brain.
- *Postsynaptically*, these opioids *open potassium channels* causing potassium to flow out of cells which makes the neurons less excitable to fire(reduced nociceptive signal transmission).

OPIOID ANALGESICS

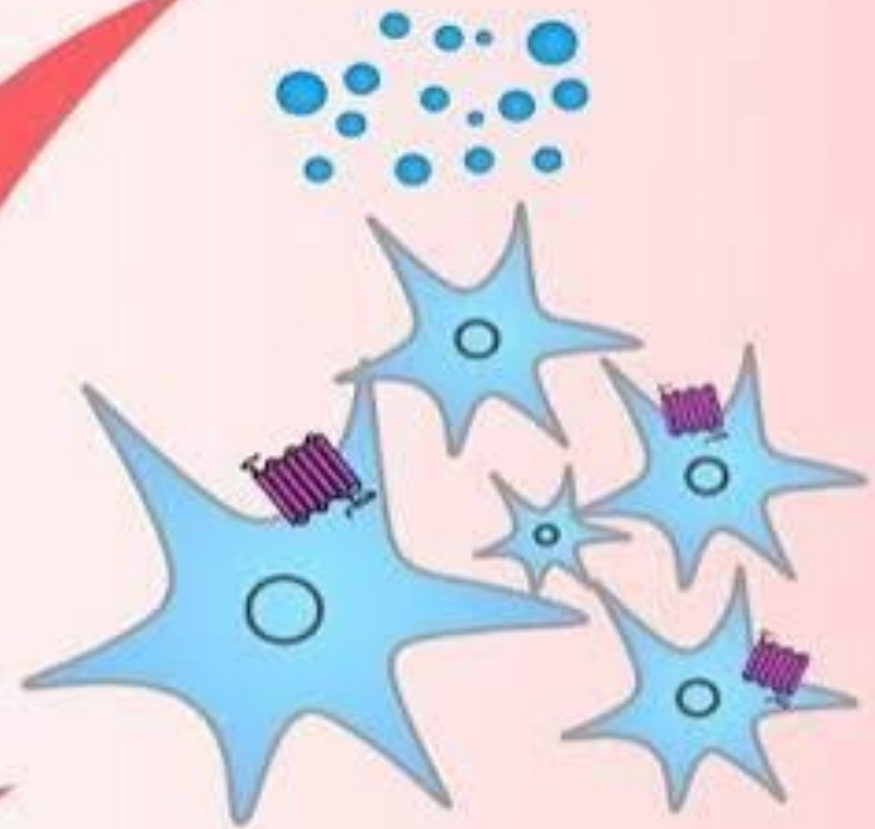
- Opioid analgesics are powerful prescription medicines used to treat moderate to severe pain that does not go away with weaker drugs.
- They work by attaching to specific receptors in the brain & body to block pain signals.
- Their actions resemble the effects of endogenous opioid system;
Decreased neurotransmitter release => decreased nociceptive neuron activity => decreased ascending pain transmission => ANALGESIA
- Common examples are *morphine, oxycodone, fentanyl, codeine & tramadol.*

Morphine

Endogenous opioids



Glutamatergic neurons



GABAergic neurons

Opioid analgesia



MOR

APPLIED PHYSIOLOGY

HYPERALGESIA

- An exaggerated response to a normally painful stimulus.
- For example, feeling intense excruciating pain when a light pinprick or minor heat stimulus touches a badly burned area of skin.

ALLODYNIA

- Pain produced by a stimulus that normally does not cause pain such as light touch.
- For example, wearing a tight shirt collar makes your skin hurt.

POST ASSESSMENT

Q1. Which structure is the major origin of descending pain inhibitory pathway?

- a) Thalamus
- b) Periaqueductal gray
- c) Cerebellum
- d) Spinal cord

Q2. Endogenous opioids reduce pain primarily by:

- a) Increasing neurotransmitter release from nociceptors
- b) Increasing neuronal excitability in the dorsal horn of the spinal cord
- c) Inhibiting neurotransmitter release & hyperpolarization of neurons
- d) Activating peripheral nociceptors

Cont..

Q3. Which statement best describes descending pain modulation?

- a) Pain signals are transmitted unchanged to the cortex
- b) Descending pathways facilitate pain transmission
- c) Descending pathways only operate during sleep
- d) Higher brain centers can enhance or inhibit nociceptive transmission

Q4. A patient experiences reduced pain during a stressful emergency despite sustaining an injury. This phenomenon is best explained by:

- a) Peripheral sensitization
- b) Stress-induced analgesia
- c) Loss of nociceptors
- d) Increased substance P release

Thank You

Any Questions?

